

Artificial Neural Network based Object Tracking

B. Resko^{1,3}, P. T. Szemes^{1,4}, P. Korondi^{1,2}, P. Baranyi^{1,3}, H. Hashimoto^{1,4}

¹ Integrated Intelligent Systems Japanese-Hungarian Laboratory

² Budapest University of Technology and Economics, Budapest, Hungary

³ Computer and Automation Research Institute of the Hungarian Academy of Sciences

⁴ University of Tokyo, Tokyo, Japan

rbarna@datatrans.hu

szemes@hlab.iis.u-tokyo.ac.jp

Abstract: Intelligent Space is a space which has distributed sensory intelligence and is equipped with actuators. The various devices of sensory intelligence cooperate with each other autonomously, and the whole space has high intelligence, where we can easily interact with computers and robots, and get useful services from them. The most important sensors in Intelligent Space are video cameras. This paper presents an artificial neural network for tracking objects moving in front of a camera. The task is performed by a biologically inspired artificial neural network, which is a first step toward cognitive systems and vision. An adaptive neural based image memory and a neural matching system is used. The proposed neural structure can be embedded in an analog neural hardware to produce a real time solution for camera motion control in object tracking that is faster and more robust than existing solutions.

Keywords: Intelligent Space, Object tracking, Computer Vision, Artificial Neural Network

1. Introduction

Intelligent Space is a space, which has distributed sensory intelligence and it is equipped with actuators¹⁾. The key concept Distributed Intelligent Network Device (DIND) is consisting of three basic elements. These are the sensor, the processor (computer) and the communication device. In the space, DIND-s monitor the space and the data acquired are shared through the network (Figure 1). Each DIND in the Intelligent Space knows its role and can support human beings. Conventionally, there is a trend to increase the intelligence of a computer to perform more and more complicated tasks or of a robot (agent) operating in a limited area. The Intelligent Space concept is the opposite of this trend. The surrounding space has ubiquitous computing and sensory intelligence instead of highly intelligent agents and a concentrated high performance computer. The most important source of information in Intelligent Space are cameras. They provide a dynamic image from multiple views within the space. If the cameras track a moving object, the object will stay in sight, and a 3D model of the space can be reconstructed.

In this paper we propose a vision system capable of tracking objects that move in front of the camera. The proposed system can be connected to a network and become the member of the DIND community within the Intelligent Space. Connecting several DIND instances of the proposed system allow a higher level DIND to calculate 3D positions within the Intelligent Space, and share the results towards other DINDs.

The heart of the vision system is a pure artificial neural network (ANN) that remembers the object that is tracked. An ANN is also used to determine the direction in which the cameras have to be rotated in order

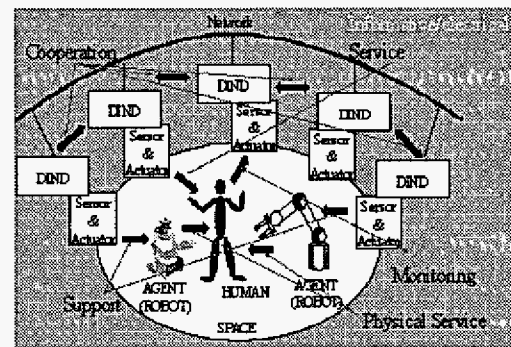


Figure 1: Intelligent Space Concept

to position the tracked object on the optical axis of the camera. The proposed ANN was inspired by the human vision system, where the image from the eyes arrive via the optic nerves into the brain. In the brain the optic nerves are crossed in a complex neural network called the optic chiasm, the output of which controls the movements of the eyes to envision the same spot. This mechanism is used when the eyes achieve stereo vision⁸⁾. The proposed system is considered to be a first step towards cognitive systems and cognitive vision. If the image coming from one eye is replaced by a neural based image memory, the same stereo vision system can be used to track the object stored in the memory. The image memory is updated with an appropriate algorithm using learning factor and the system is able to adapt to changes in the tracked object. The main contribution of this paper is a pure ANN that receives two image parts, one from a camera and one from the neural based image memory. On the output

it provides a control signal to the actuators moving the camera, just like the human brain to the muscles that move the eyes. The proposed ANN does not require any conventional computational tools (PC). The network is constructed of neurons performing very simple tasks, such as additions, subtractions, etc. For this reason the input weights and the features of the transfer functions can be pre-determined, thus no learning process is required. Preset input weights is not a novelty, preset ANN has been used as a fast multiplier³⁾. The ANN implemented in a neural hardware using a neural memory and an analogue neural network it solves the tracking problem in constant time. In the lack of such a hardware, a PC based simulation has been used to test and evaluate the capabilities of the system.

The organization of this paper is as follows. In Section 2 a brief introduction about the image features used in the matching process is given. In Section 3 the proposed neural system is described including the matching of pixel pairs, the proposed artificial neural network, the neural memory and the preadjustment of input weights and transfer functions. The possible hardware implementation of the system will be treated here. In Section 4 test results are presented, and finally in section 5 conclusions with a discussion of the results is made and further directions of the development is presented.

2. Image features

In the presented system three pixel properties are used in the comparisons to find the corresponding pixels on the two images acquired from either the image memory or the camera⁷⁾. These pixel properties are intensity, gradient magnitude and gradient orientation (I, D, O respectively) Figure 2. Denote the intensity of an arbitrary pixel in location (x, y) as $I = f(x, y)$, the gradient of $f(x, y)$ is defined as

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} \quad (1)$$

The magnitude of the gradient is defined as the variation of $f(x, y)$ and is written as

$$D = \|\nabla f\| = \sqrt{G_x^2 + G_y^2} \quad (2)$$

The gradient orientation $\alpha(x, y)$ of the vector ∇f is defined in location (x, y) as

$$O = \alpha(x, y) = \tan^{-1} \left(\frac{G_x}{G_y} \right) \quad (3)$$

3. The Proposed System for Object Tracking

3.1 Pixel Pair Matching

The goal of the proposed system is to adjust the optical axis of the camera on the object stored in the image

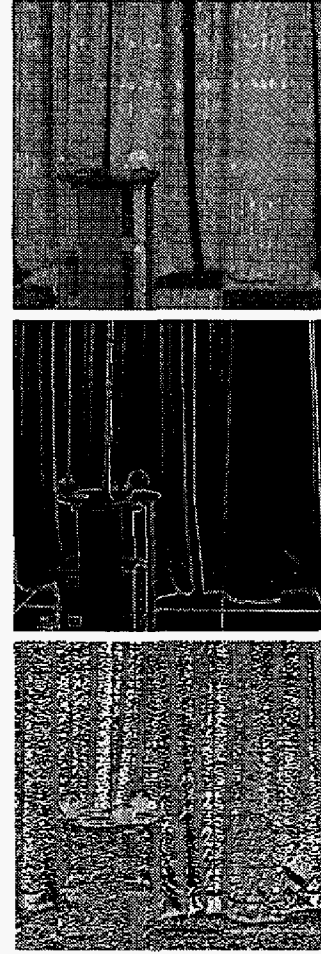


Figure 2: The intensity (top), difference (middle) and orientation (bottom) values of the pixels of an image

memory. The human eyes also look at the spot of interest in order to get the most information about it. The following assumptions were made in the choice of the pixels to be compared with each other.

The images are supposed to be in a master-slave relationship. The master image is the one that is stored in the image memory, and it serves as a target image. The slave image is arriving from the camera. The system adjusts the optical center of the camera that is providing the slave image so that the same point P is projected on the optical center of the camera as in the middle of the image memory. A window of 9×9 pixels is considered as a pixel surrounding. Thus on the master image there is a single pixel with one window. The final task is to find a window on the slave image that best matches the window of the master image.

Not all the pixels on the slave image are taken into consideration. In our case the camera can only be rotated horizontally. As a result all the 9×9 -pixel-sized windows on the horizontal line of the slave image are compared to the single window in the center of the mas-



Figure 3: Image parts feed into the ANN. Master image (left) and Slave image (right)

ter image, thus the interesting image parts are the 9×9 square on the master image and an $X_{max} \times 9$ band on the slave image, where X_{max} is the width of the slave image in pixels. (Figure 3)

A color image can be decomposed to three images according to the RGB color components. In our system 8-bit gray scale images are used. Because the intensity value of each pixel is sensitive to changes in contrast and illumination, comparing only the intensities of the pixels and their surroundings to acquire a matching degree does not yield acceptable results. The gradient of each pixel is also taken into consideration during comparison of two pixel surroundings. We suppose that three measures for each pixel are known, which are intensity, difference, orientation. Denote I_M^p and I_S^p the intensity of the master and slave pixels being matched, D_M^p and D_S^p the differential value of the master and slave pixels being matched and O_M^p and O_S^p the gradient orientation of the master and slave pixels being matched. The matching degree of a pixel pair $p_M(x_M, y_M)$ and $p_S(x_S, y_S)$ is defined as the average of the absolute difference between the corresponding feature values of the pixels. Denote $d(p_M, p_S)$ as the matching degree, it can be written in the following form:

$$d(p_M, p_S) = \frac{|I_M^p - I_S^p| + |D_M^p - D_S^p| + |O_M^p - O_S^p|}{3} \quad (4)$$

This choice has been made because it gives a true

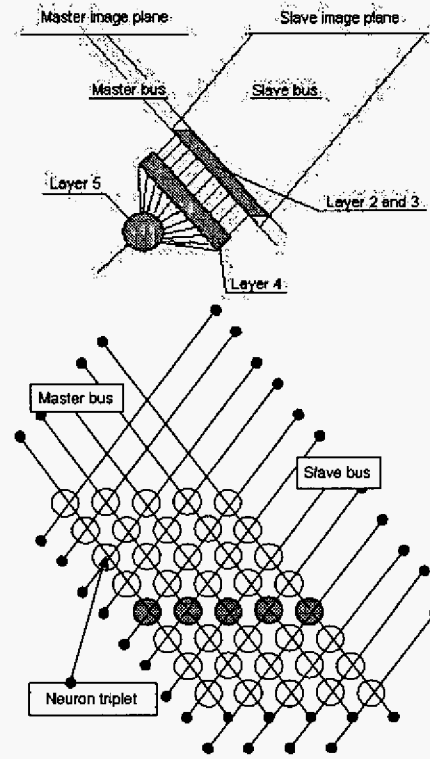


Figure 4: The overview of the proposed ANN (top) and a section of the 3D array holding the neurons of the second layer (bottom). Each line represents the difference of one window pair.

degree of similarity between pixels, it can be computed by an ANN using input weights that are easy to adjust in advance.

A neural based image memory is also used in the system. Such a memory overwrites old images with a combination of the old image and the new image. Denote $\mathcal{F}_{old}$ the image stored in the image memory, and $\mathcal{F}_{new}$ the image arriving from the camera. The next image that will be stored in the image memory will be

$$(1 - \lambda)\mathcal{F}_{old} + \lambda\mathcal{F}_{new}, \quad (5)$$

where λ is a constant training parameter.

In the next section the proposed neural system will be presented along with the description of the chosen input weights and transfer functions.

3.2 Neural Based System for Pixel Matching and Camera Control

The structure of the proposed neural system is shown on Figure 4. The image arriving from the camera is transmitted both to the ANN and the image memory. The image memory updates itself using the camera image according to equation 5 if the control signal from the ANN is *neutral*, meaning that the camera is looking in the right direction. Otherwise the camera image is neglected by the image memory.

The ANN receives two images, one from the image memory and one from the camera. After comparing the two images, it provides a control signal on its output, which has two destinations. One destination is the image memory, the other one is the camera moving mechanism. If the control signal is neutral for the camera moving mechanism, then the image memory is updated with the current image from the camera according to equation 5.

In the rest of this section the ANN to compare the pixel surroundings will be explained in detail.

The proposed neural structure is shown on Figure 4 (top). The system is composed of 5 layers. Each layer is responsible for a different task (input, matching degree, decision, output), which yields a clear structure that is easy to understand and further develop. In the first layer we suppose to have the input data from the interesting image regions, i.e. the 9×9 square on the center of the master image and the $X_{max} \times 9$ band from the slave image. These data are feeded into the input layer of the ANN.

The outputs of the input neurons form two buses, a master and a slave bus, as an analogy of the optic nerves behind the human eyes. The master bus M for each pixel p of the 9×9 window on the master image carries the I_M^p, D_M^p and O_M^p values, thus groups $3 \times 9 \times 9 = 243$ outputs of the input layer. Let $M_{i,j}$ denote the I_M^p, D_M^p and O_M^p values that belong to the pixel $p(i,j)$ in the master window. The slave bus S carries the I_S^p, D_S^p and O_S^p values of each pixel on the $X_{max} \times 9$ band on the slave image, thus groups $3 \times 9 \times X_{max}$ outputs of the input layer. Let $S_{k,j}$ denote the I_S^p, D_S^p and O_S^p values that belong to the pixel $p(k,j)$ in the band on the slave image. In the next layer the two buses are crossed in a 3-dimensional array A of $9 \times 9 \times (X_{max} - 8)$ neuron triplets. This builds up the second layer of the proposed ANN. A two dimensional section where j is set to a constant value is shown in Figure 4 (bottom).

In array A each neuron triplet calculates the absolute difference of the $I_M^p, I_S^p, D_M^p, D_S^p$ and O_M^p, O_S^p values with respect to the pixels from which their inputs are arriving from.

$$A_{i,k,j}^{(I)} = |M_{i,j}^{(I)} - S_{i+k,j}^{(I)}|, \quad (6)$$

$$A_{i,k,j}^{(D)} = |M_{i,j}^{(D)} - S_{i+k,j}^{(D)}|, \quad (7)$$

$$A_{i,k,j}^{(O)} = |M_{i,j}^{(O)} - S_{i+k,j}^{(O)}|, \quad (8)$$

$i, j \in [1..9], k \in [1..X_{max} - 8],$

where for example $M_{i,j}^{(I)}$ denotes the intensity value arriving from the $p(i,j)$ pixel of the master image.

The third layer neurons of the network are situated in a similar 3D array as those in the second layer. The difference is that in the third layer there is only one neuron in each cell of the array. The outputs of the three corresponding neurons in layer 2 are connected to the inputs of the neurons in layer 3. Denote the array containing the third-layer neurons B , then the

operation performed by a third layer neuron is described as

$$B_{i,k,j} = \frac{A_{i,k,j}^{(I)} + A_{i,k,j}^{(D)} + A_{i,k,j}^{(O)}}{3} \quad (9)$$

This means that the third layer neuron simply calculates the mean value of the outputs coming from the three second-layer neurons. The output is then the same as it was written in Equation 4. In other words, $B_{i,k,j}$ equals the mean difference between the pixels $p(i,j)$ of the master image window and $p(k+i,j)$ of the slave image band. It is to note that if k is increased by one, then B will contain the mean difference between the same pixel on the master image and the next pixel to the right on the slave image.

To compute the matching degree between two pixels, their surroundings also have to be taken into consideration. The array $B_{i,c,j}$ where c is a constant, contains the average difference between the 81 pixels in the 9×9 master window and the c^{th} slave window. To get the matching degree of a pixel pair, simply the mean value of the 81 difference values is calculated, and subtracted from 255, the maximum value of the difference. This is done by the fourth layer of the network, which contains $X_{max} - 8$ neurons with 81 inputs for each of them. The output of the k^{th} neuron equals to the matching degree of the master window and the k^{th} window on the slave image.

The original goal was to adjust the optical center of the camera on the point P the surrounding of which is stored in the image memory. The goal can be considered to be achieved when the highest matching degree is given by the $X_{max} - 8/2^{th}$ neuron. In other words the highest matching has to occur in the central neuron of the fourth layer.

It is the task of the fifth layer to decide if the highest matching value is in the middle of the fourth layer, and if not, then in which direction from the center it is situated. In the fifth layer there is only one special neuron, which outputs -1, 0 or 1 according to the input values. The matching degree values from the previous layer are fed into this special neuron. If the maximum of the inputs is situated to the right from the central input of the neuron, it outputs -1, while if the maximal value is to the left, it outputs 1. If the maximal input is the central input itself, the output will be 0. Finally a simple analog hardware tells the servo-motor mechanism to drive the camera to the appropriate direction in order to achieve an optimal matching.

4. Testing and Evaluation

To evaluate the proposed system, a mechanism was built with a camera mounted on the top. A mechanism was adjusting the orientation of the camera. The ANN was simulated on a PC.

An interesting result is the matching degree between the target image and the various image windows situated on the slave image band.

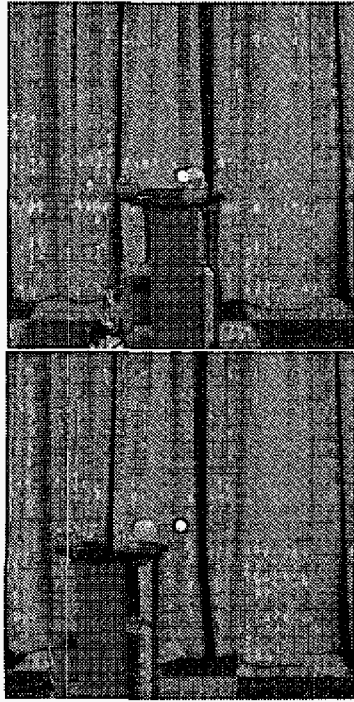


Figure 5: Master image (top) and slave image (bottom). The image centers are indicated by the white dots on both images. The matching pixel on the slave image is located to the left from the center of the slave image.

Bellow an image pair is presented with the matching degree graph and the direction suggested by the system.

On the presented pair (Figure 5) of images the center of the master image (top) corresponds to a pixel that is situated to the left from the center of the slave image (bottom). The matching degrees are shown on Figure 6. In this case the peak is to the left from the center, meaning that the slave camera has to be rotated to the left.

Using a real-time simulation of the system, the camera was able to follow objects like a human face. To start the test, the human had to be still, while the system has learnt the face. Afterwards the face could be moved,

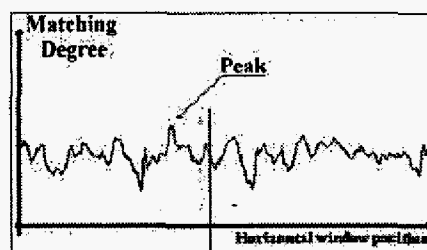


Figure 6: The matching degree of the image pair. The peak representing the best match is to the left from the center.

and the camera was following it. When the camera has reached the desired orientation, the image memory was updated. This allows slow changes in the object being followed, and makes the system robust against noise and other perturbations.

5. Hardware implementation

The main contribution of this paper is a pure ANN that receives two image parts, one from a camera and one from the neural based image memory. On the output it provides a control signal to the actuators moving the camera, just like the human brain to the muscles that move the eyes.

The proposed ANN does not require any conventional computational tools (PC). Implemented in a neural hardware using a neural memory and an analogue neural network it solves the tracking problem in constant time. In the lack of such a hardware, a PC based simulation has been used to test and evaluate the capabilities of the system.

6. Conclusion

The proposed system provides a fast and robust method for object tracking. The input uses not only pixel intensities, but also image features to track the object. This yields a more precise and robust tracking algorithms than those used in other camera applications. In the future it will be embedded in a robot, or it can be used to track human faces in Intelligent Space, even if they turn around.

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